AN AUGMENTATION TOWARDS D INDUCES MORPHISMS OF TOPOI A_N AND A ON THE TOTAL SITE WITH A_* COMPUTED AS AN EQUALIZER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: item (3) holds by the definition of a_* as the displayed equalizer (source lines 1204-1214), and the relation a g_n = a_n (line 1217, printed 'f_n' in the source, a typo for a_n) is the one-line componentwise check (a g_n)^{-1}G = g_n^{-1}(a^{-1}G) = (a^{-1}G)_n = a_n^{-1}G using the componentwise description of a^{-1} and part (1) of the restriction-to-components lemma; the author's spellings (including 'coosimplicial' and 'adjuntion') are preserved verbatim; this lemma supplies the equalizer description used by cards L46151 and the chapter's descent arguments.

Editorial note. The label lemma-augmentation-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The augmentation datum of the simplicial site. Same chapter, Remark \label{remark-augmentation-site} and Situation \ref{situation-simplicial-site}. Lines 1142-1145: the functors u_0, u_varphi with the relations u_varphi u_0-style compositions of the datum.
- Sites, composition of morphisms of sites (case A). Sites chapter, \label{sites-lemma-composition-morphisms-sites}. Lines 1163-1169: u_n = u_varphi u_0 defines the morphism of sites a_n and gives a_m f_varphi = a_n.
- Sites, composition of cocontinuous functors (case B). Sites chapter, \label{sites-lemma-composition-cocontinuous}. Lines 1171-1177: the cocontinuous u_n defines a_n : Sh(C_n) -> Sh(D) with the same compatibilities.
- Sheaves on the total site described via components and transitions. Same chapter, \label{lemma-describe-sheaves-simplicial-site-site}. Lines 1189-1195: the componentwise definition of a^{-1}G with transition maps the identifications f_varphi^{-1} a_m^{-1}G = a_n^{-1}G.
- The cosimplicial augmentation lemma for maps into cosimplicial objects. Simplicial chapter, \label{simplicial-lemma-augmentation-howto} (dual). Lines 1222-1226: maps G_bullet -> a_{bullet,*}F of cosimplicial sheaves are the same as maps G -> a_*F.

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- The component morphisms g_n of the total site. Same chapter, \label{lemma-restriction-to-components-site} (card L46149 of this batch). Lines 1152-1155: item (2) of the statement references these g_n; a g_n = a_n is checked from the componentwise description.

Source reference. Original reference 09WF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at proof line(s) 1193; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1154; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6Z: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1125–1125. Prerequisite pointer invoked at proof line(s) 1144; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 018H: sources/stacks-chow/tags.txt, source lines 1616–1616. Prerequisite pointer invoked at proof line(s) 1225; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03L5: sources/stacks-chow/tags.txt, source lines 4701–4701. Prerequisite pointer invoked at proof line(s) 1176; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03CB: sources/stacks-chow/tags.txt, source lines 4378–4378. Prerequisite pointer invoked at proof line(s) 1167; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1142; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let a_0 be an augmentation towards a site $\mathcal{D}$ as in Remark 0D6Z. Then a_0 induces*

(1) *a morphism of topoi $a_n : Sh(\mathcal{C}_n) \to Sh(\mathcal{D})$ for all $n \geq 0$,*
(2) *a morphism of topoi $a : Sh(\mathcal{C}_{total}) \to Sh(\mathcal{D})$*

such that

(1) *for all $\varphi : [m] \to [n]$ we have $a_m \circ f_\varphi = a_n$,*
(2) *if $g_n : Sh(\mathcal{C}_n) \to Sh(\mathcal{C}_{total})$ is as in Lemma 09WG, then $a \circ g_n = a_n$, and*
(3) *$a_*\mathcal{F}$ for $\mathcal{F} \in Sh(\mathcal{C}_{total})$ is the equalizer of the two maps $a_{0,*}\mathcal{F}_0 \to a_{1,*}\mathcal{F}_1$.*

Proof. Case A. Let $u_n : \mathcal{D} \to \mathcal{C}_n$ be the common value of the functors $u_\varphi \circ u_0$ for $\varphi : [0] \to [n]$. Then u_n corresponds to a morphism of sites $a_n : \mathcal{C}_n \to \mathcal{D}$, see Sites, Lemma 03CB. The same lemma shows that for all $\varphi : [m] \to [n]$ we have $a_m \circ f_\varphi = a_n$.

Case B. Let $u_n : \mathcal{C}_n \to \mathcal{D}$ be the common value of the functors $u_0 \circ u_\varphi$ for $\varphi : [0] \to [n]$. Then u_n is cocontinuous and hence defines a morphism of topoi $a_n : Sh(\mathcal{C}_n) \to Sh(\mathcal{D})$, see Sites, Lemma 03L5. The same lemma shows that for all $\varphi : [m] \to [n]$ we have $a_m \circ f_\varphi = a_n$.

Consider the functor $a^{-1} : Sh(\mathcal{D}) \to Sh(\mathcal{C}_{total})$ which to a sheaf of sets $\mathcal{G}$ associates the sheaf $\mathcal{F} = a^{-1}\mathcal{G}$ whose components are $a_n^{-1}\mathcal{G}$ and whose transition maps $\mathcal{F}(\varphi)$ are the identifications

$$f_\varphi^{-1}\mathcal{F}_m = f_\varphi^{-1}a_m^{-1}\mathcal{G} = a_n^{-1}\mathcal{G} = \mathcal{F}_n$$

for $\varphi : [m] \to [n]$, see the description of $Sh(\mathcal{C}_{total})$ in Lemma 09WF. Since the functors a_n^{-1} are exact, a^{-1} is an exact functor. Finally, for $a_* : Sh(\mathcal{C}_{total}) \to Sh(\mathcal{D})$ we take the functor which to a sheaf $\mathcal{F}$ on $Sh(\mathcal{D})$ associates

$$a_*\mathcal{F} = \text{Equalizer}(a_{0,*}\mathcal{F}_0 \rightrightarrows a_{1,*}\mathcal{F}_1)$$

Here the two maps come from the two maps $\varphi : [0] \to [1]$ via

$$a_{0,*}\mathcal{F}_0 \to a_{0,*}f_{\varphi,*}f_\varphi^{-1}\mathcal{F}_0 \xrightarrow{\mathcal{F}(\varphi)} a_{0,*}f_{\varphi,*}\mathcal{F}_1 = a_{1,*}\mathcal{F}_1$$

where the first arrow comes from $1 \to f_{\varphi,*}f_\varphi^{-1}$. Let $\mathcal{G}_\bullet$ denote the constant coosimplicial sheaf with value $\mathcal{G}$ and let $a_{\bullet,*}\mathcal{F}$ denote the cosimplicial sheaf having $a_{n,*}\mathcal{F}_n$ in degree n. By the usual adjuntion for the morphisms of topoi a_n we see that a map $a^{-1}\mathcal{G} \to \mathcal{F}$ is the same thing as a map

$$\mathcal{G}_\bullet \longrightarrow a_{\bullet,*}\mathcal{F}$$

of cosimplicial sheaves. By the dual to Simplicial, Lemma 018H this is the same thing as a map $\mathcal{G} \to a_*\mathcal{F}$. Thus a^{-1} and a_* are adjoint functors and we obtain our morphism of topoi a[1]. The equalities $a \circ g_n = f_n$ follow immediately from the definitions. □

[1]In case B the morphism a corresponds to the cocontinuous functor $\mathcal{C}_{total} \to \mathcal{D}$ sending U in $\mathcal{C}_n$ to $u_n(U)$.